microring modulators.

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- W-Shaped PN junction.
- Lateral PN junction simulation.
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contents

- W-Shaped PN junction.

W-Shaped PN junction

- *The paper proposes a **W-shaped PN doping profile** for a silicon microring which **boosts modulation efficiency ≈2.3×** versus a standard lateral PN at the same doping, producing much larger wavelength shift per volt, higher, while keeping losses manageable — at the cost of increased device capacitance and by so BW degradation.

*Hamouda, Mahmoud, et al. "High-efficiency PAM4 ring modulator using W-Shaped PN junction." Optical Interconnects and Packaging 2025. Vol. 13372. SPIE, 2025.

W-Shaped PN junction

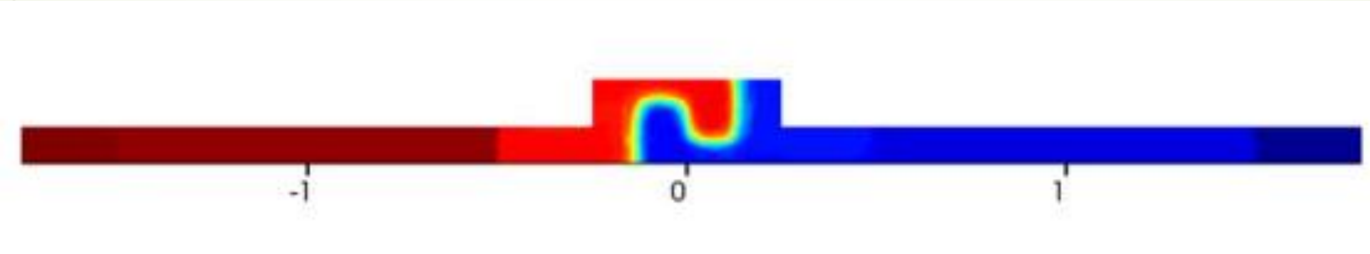
- What the W-shaped junction is?

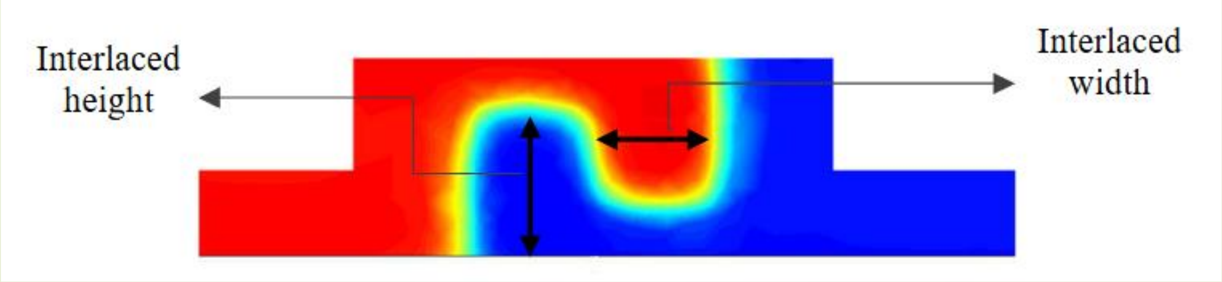
→ A patterned doping cross-section shaped like a "W" (two side lobes and a center region) that **increases the depletion-region overlap with the optical mode** more effectively than a simple lateral PN.

More depletion **overlap** hence increased **interaction** between the optical mode and the free carriers → **more Δn per volt** → larger resonance shift per V (higher pm/V, lower $V\pi \cdot L$).

W-Shaped PN junction

- What the W-shaped junction is?





W-Shaped PN junction

- **Key parameters**

Doping used: $Ne = Nh = \mathbf{5 \times 10^{18}}$ cm−3.

Wavelength shift / V: W-shaped ≈ **163** pm/V (≈**2.3×**):

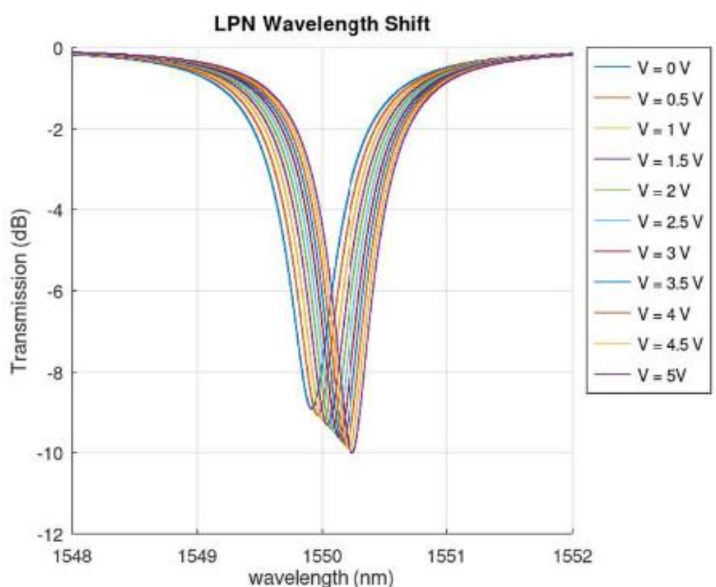
lateral ≈ **70** pm/V.

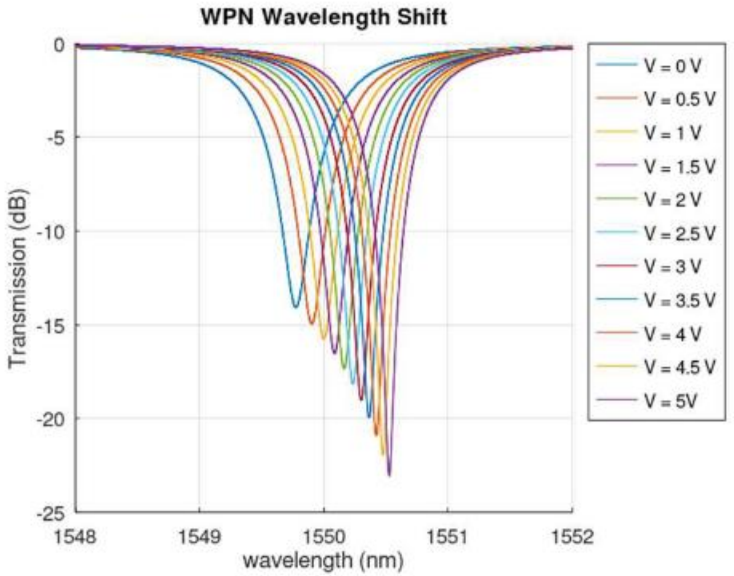
Vπ ·L (@ Vbias = −2 V): W-shaped **0.24** V ·cm vs lateral **0.59** V ·cm.

Loss (dB/mm @ −2 V): W-shaped **12.6** dB/mm vs lateral **15.2** dB/mm.

Capacitance (@ −2 V): W-shaped **1.6** fF/cm vs lateral **0.7** fF/cm

W-Shaped PN junction



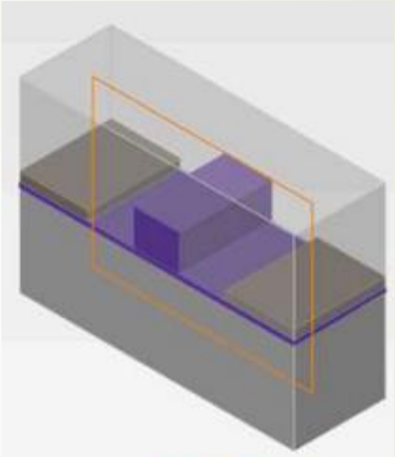


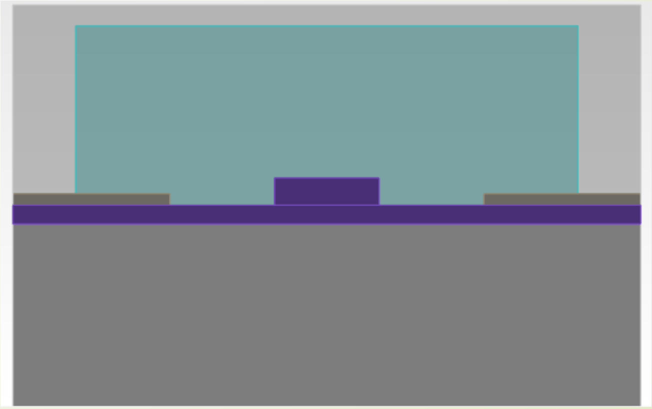
contents

- Lateral PN junction simulation.

Lateral PN junction simulation

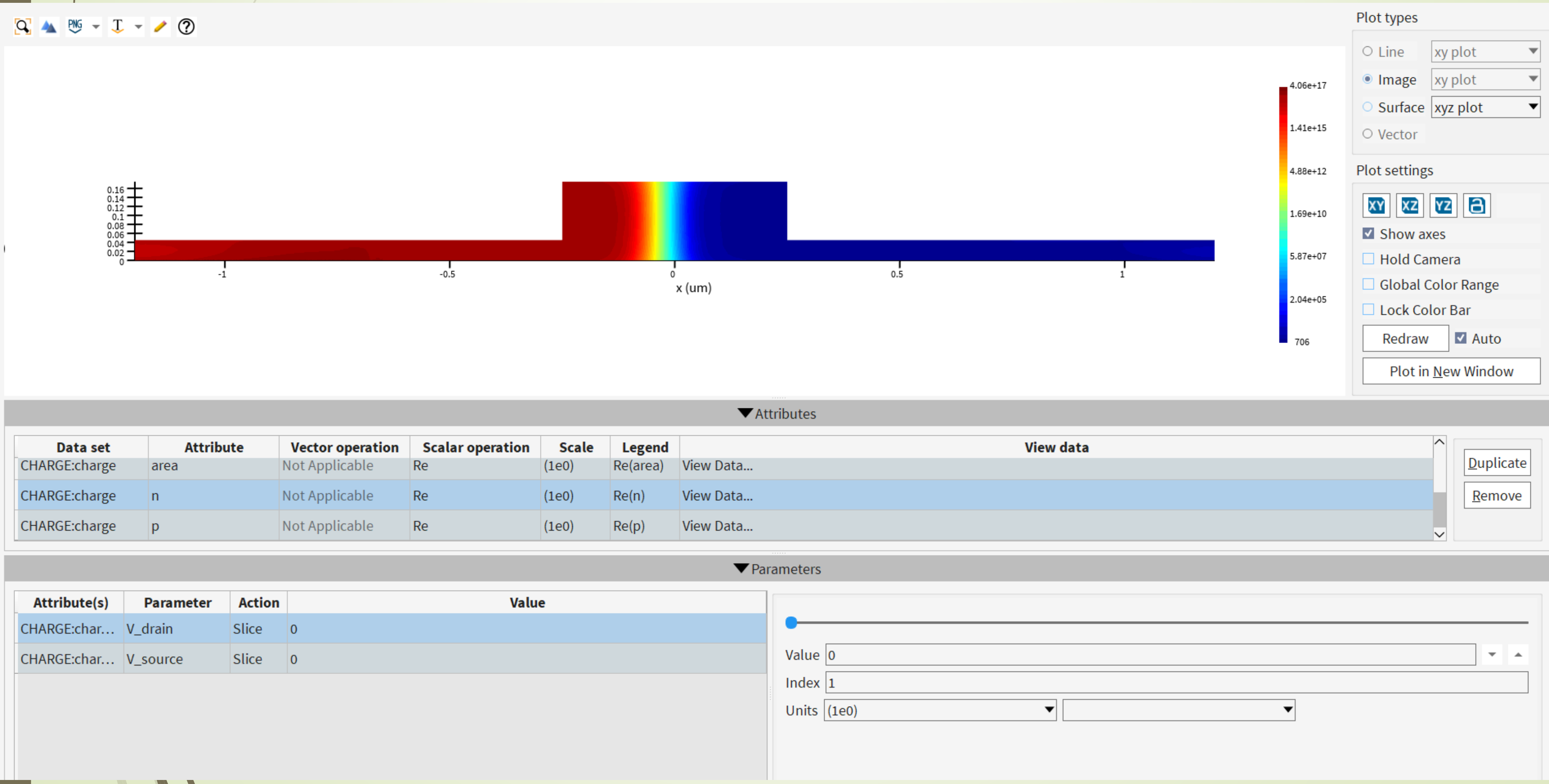
- The effect of bias voltage on carriers' concentration.
- Plot capacitance vs Voltage.
- Mode profiles
- Δn and α





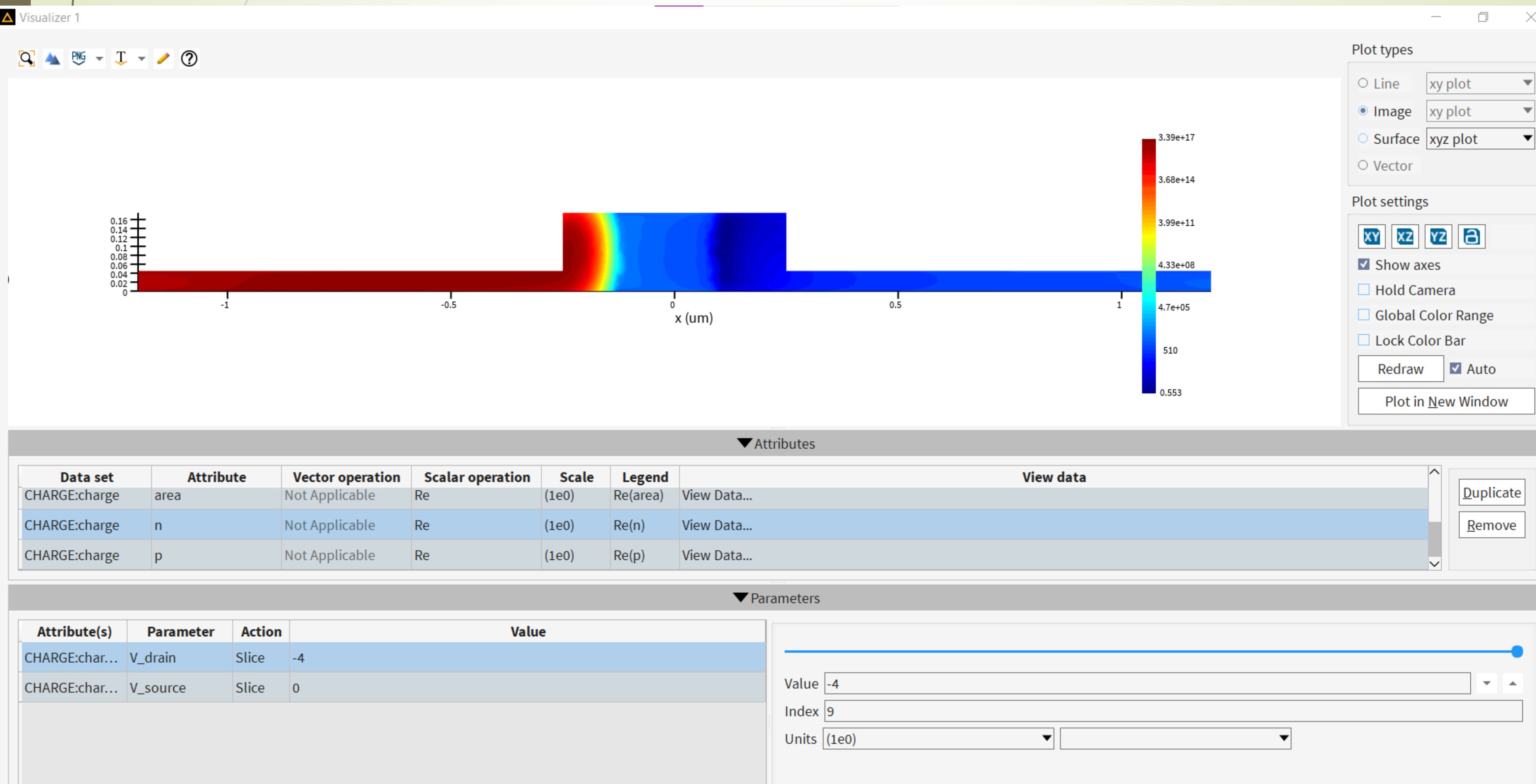
Lateral PN junction simulation.

Charge Distribution Profile (log scale)



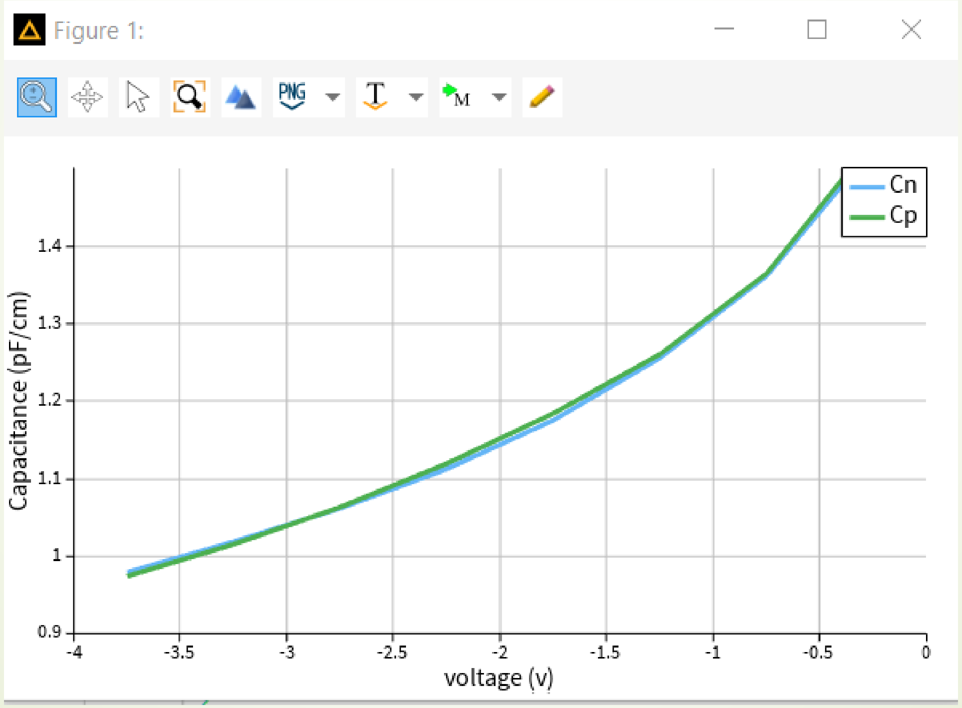
Lateral PN junction simulation.

Charge Distribution Profile (log scale)



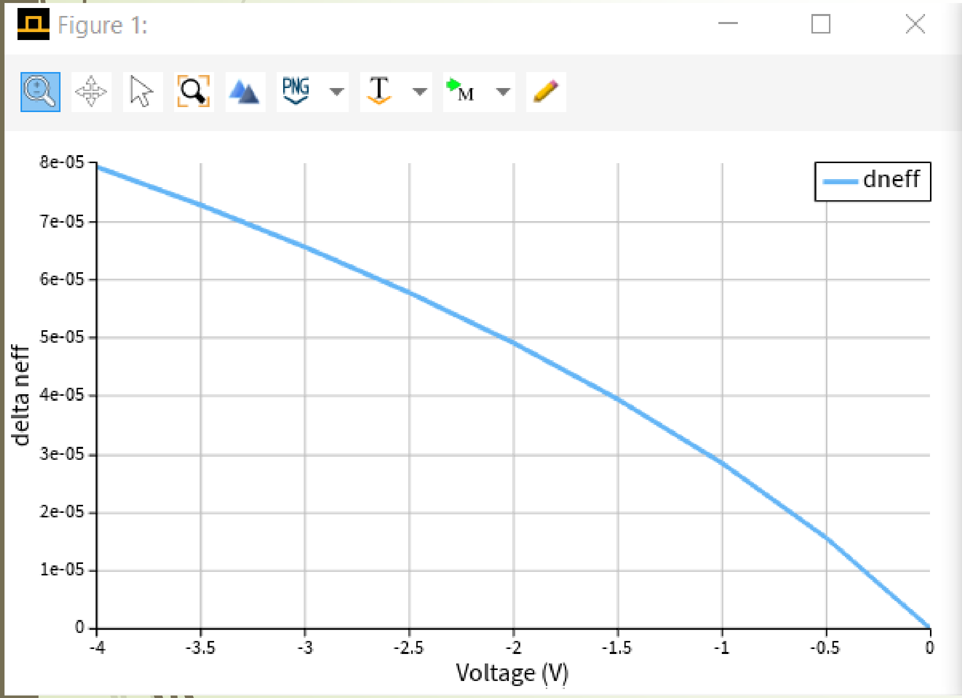
Lateral PN junction simulation.

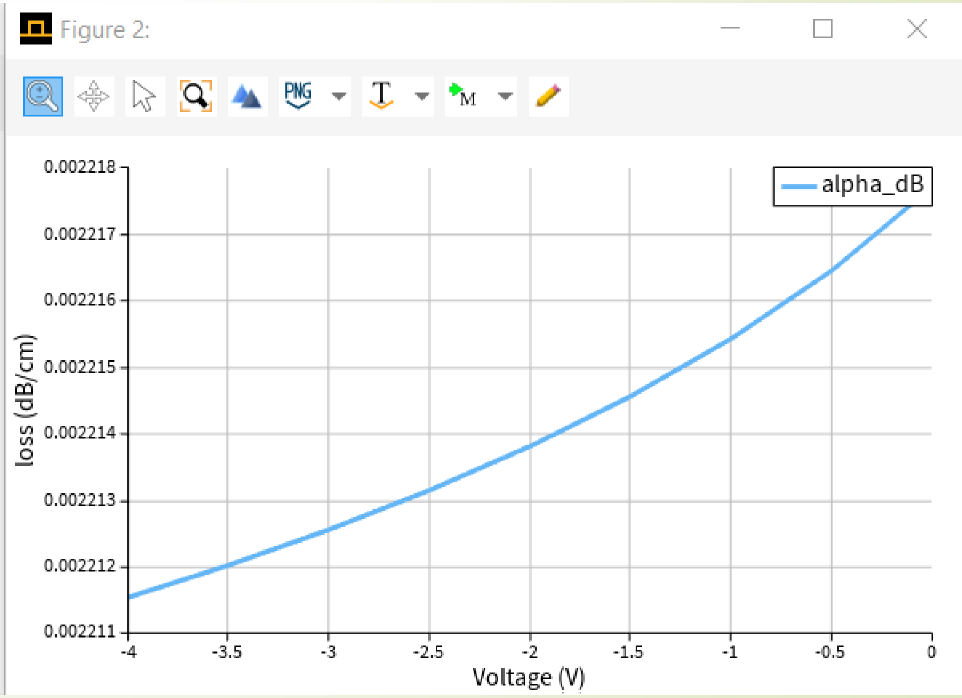
- **C-V plot of the device**



Lateral PN junction simulation.

- **Δn and α plots of the device**





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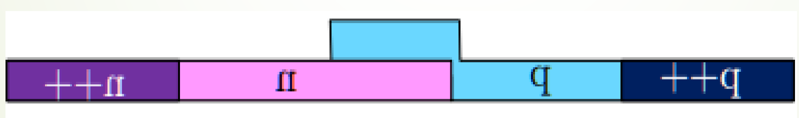
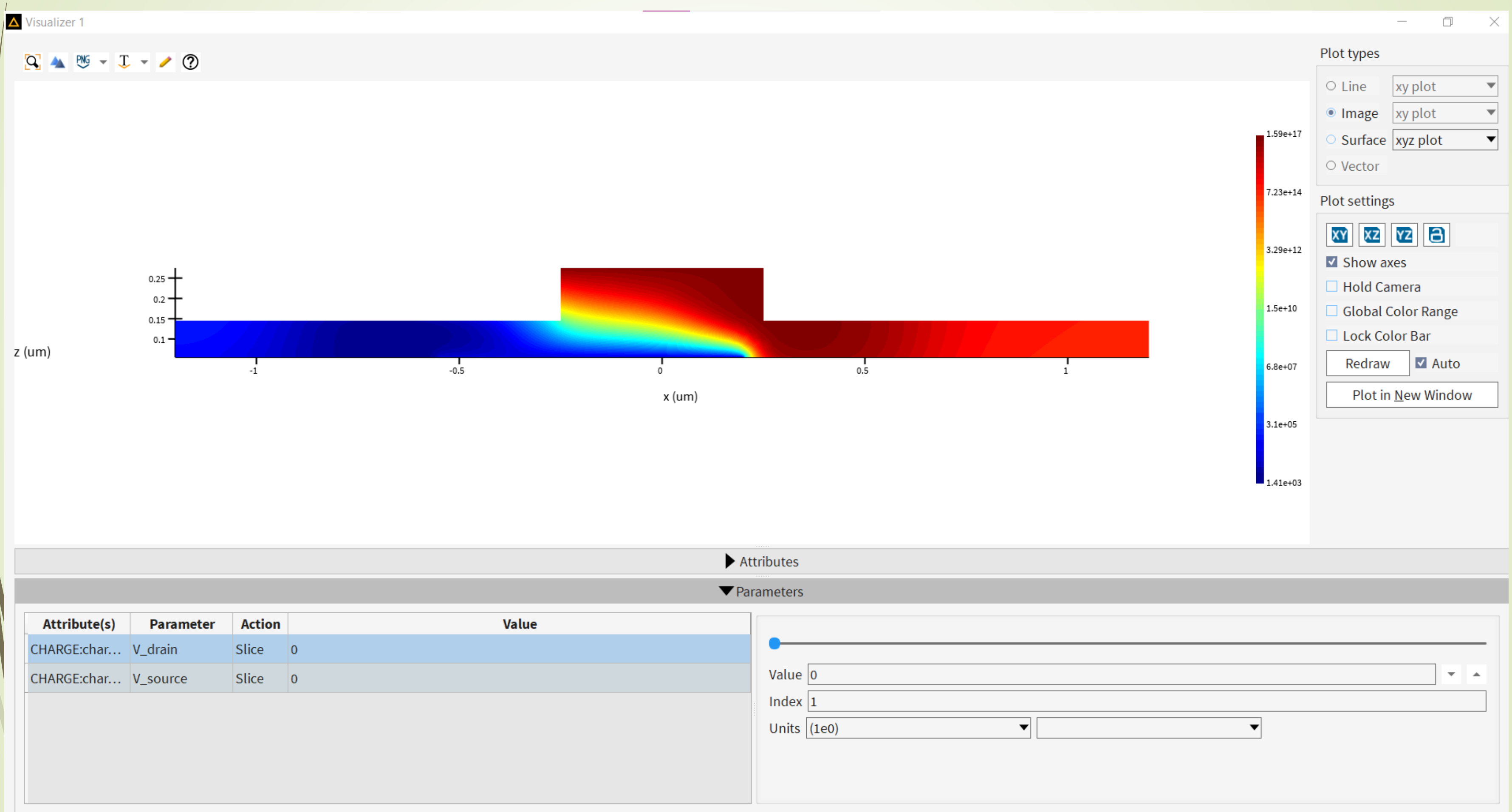
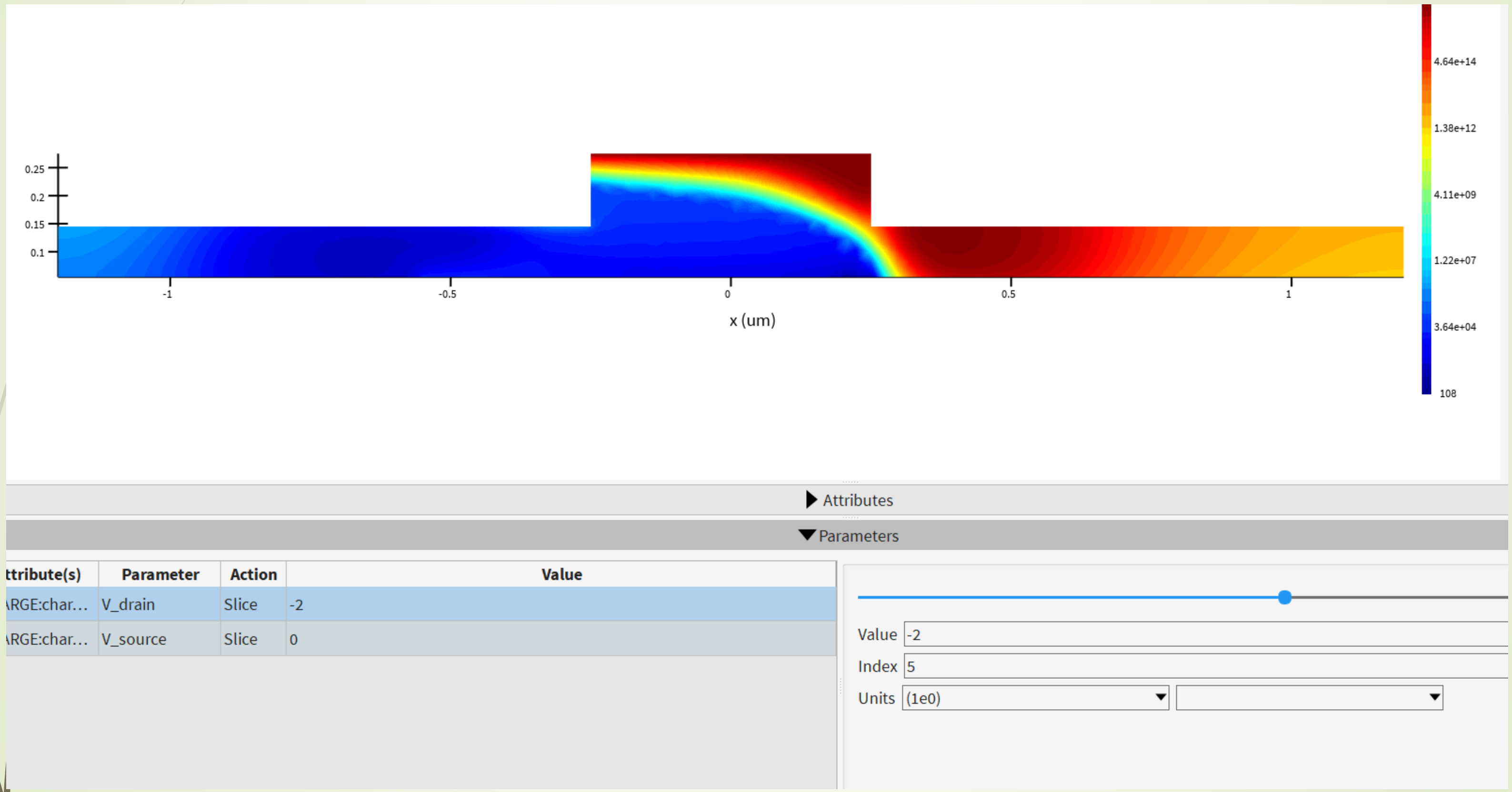
- Vertical PN junction simulation.



Vertical PN junction simulation

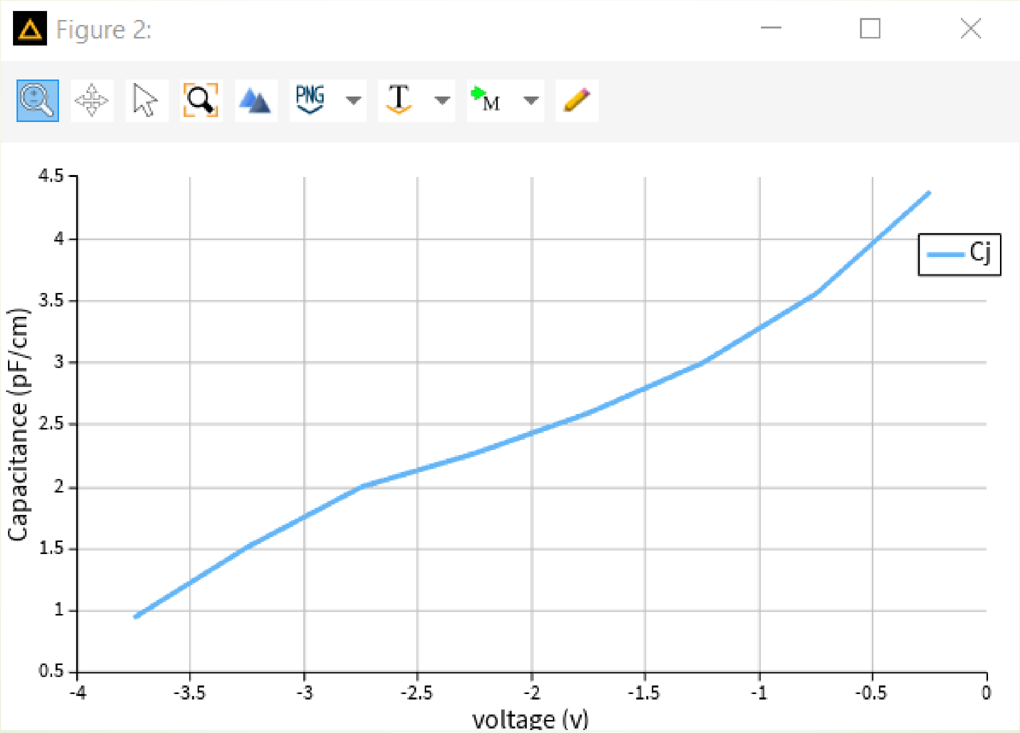


Vertical PN junction simulation



Vertical PN junction simulation

C-V plot of the device



Thanks ☺